

SELECTED
PUBLICATIONS

Bellec*, G., **Scherr*, F.**, Subramoney, A., Hajek, E., Salaj, D., Legenstein, R., & Maass, W.
A solution to the learning dilemma for recurrent networks of spiking neurons.
Nature Communications (2020)

Scherr, F., Stöckl, C., & Maass, W.
One-shot learning with spiking neural networks.
Submitted (2020)

Bellec*, G., **Scherr*, F.**, Hajek, E., Salaj, D., Subramoney, A., Legenstein, R., & Maass, W.
Eligibility Traces provide a data-inspired alternative to backpropagation through time
NeurIPS workshop on Real neurons and hidden units (Talk) (2019)

Subramoney*, A., Bellec*, G., **Scherr*, F.**, Hajek, E., Salaj, D., Legenstein, R., & Maass, W.
Slow processes of neurons enable a biologically plausible approximation to policy gradient
NeurIPS workshop on biological and artificial RL (Talk) (2019)

OTHER
PUBLICATIONS

Zenke, F., Bohté, S. M., Clopath, C., Comşa, I. M., Gölitz, J., Maass, W., Masquelier, T., Naud, R., Neftci, E. O., Petrovici, M. A., **Scherr, F.**, & Goodman, D. F.

Visualizing a joint future of neuroscience and neuromorphic engineering
Neuron (2021)

Scherr, F., & Maass, W.
Learning-to-learn for neuromorphic hardware.
To appear in Neuromorphic Computing and Engineering, IOP Publishing (2021)

Subramoney, A., Bellec, G., **Scherr, F.**, Legenstein, R., & Maass, W.
Revisiting the role of synaptic plasticity and network dynamics for fast learning in spiking neural networks.
Submitted (2021)

Subramoney, A., **Scherr, F.**, & Maass, W.
Reservoirs learn to learn
Submitted (2019)

Bohnstingl*, T., **Scherr*, F.**, & Maass, W.
Neuromorphic hardware learns to learn
Frontiers in Neuroscience (2019)

Bellec*, G., **Scherr*, F.**, Hajek, E., Salaj, D., Legenstein, R., & Maass, W.
Biologically inspired alternatives to backpropagation through time for learning in recurrent neural nets
arXiv (2019)

* equal contribution



EDUCATION

UNIVERSITY OF TECHNOLOGY, Graz, Austria
Institute of Theoretical Computer Science

MAY 2018 - DEC 2020

PhD in Computer science (with honours, GPA 4.0/4.0)
Advisor: **Prof. Wolfgang Maass**

Thesis: Learning from rewards and with priors in
recurrent networks of artificial and spiking neurons

UNIVERSITY OF TECHNOLOGY, Graz, Austria
Institute of Theoretical Computer Science

OCT 2016 - APR 2018

MSc in Information and Computer Engineering (with honours, GPA 3.9/4.0)

Thesis: Spike-based agents for multi-armed bandits
Major: Computational Intelligence
Minor: Information Security

UNIVERSITY OF TECHNOLOGY, Graz, Austria
Institute of Theoretical Computer Science

OCT 2014 - APR 2018

BSc in Physics (with honours, GPA 3.8/4.0)

Thesis: Gradient-based optimization of AMEA parameters

UNIVERSITY OF TECHNOLOGY, Graz, Austria
Institute of Theoretical Computer Science

OCT 2013 - APR 2016

BSc in Information and Computer Engineering (with honours, GPA 3.7/4.0)

Thesis: Automated security proofs for symmetric ciphers



EXPERIENCE

ICML conference

FEB 2021

Reviewer

UNIVERSITY OF TECHNOLOGY, Graz, Austria
Institute of Theoretical Computer Science

MAY 2018 - PRESENT

Research Assistant

ANYCONCEPT, Graz, Austria

OCT 2020 - PRESENT

Academic Mentor

TREVER, Graz, Austria

JUN 2019 - PRESENT

Academic Mentor

NEURIPS conference

JUL 2020

Reviewer

UNIVERSITY OF TECHNOLOGY, Graz, Austria
Institute of Applied Information Processing and Communications

MAR 2018 - JUN 2018

Undergraduate teaching assistant in computer networks

UNIVERSITY OF TECHNOLOGY, Graz, Austria
Institute of Applied Information Processing and Communications

MAR 2017 - JUN 2017

Undergraduate teaching assistant in computer networks

UNIVERSITY OF TECHNOLOGY, Graz, Austria
Institute of Applied Information Processing and Communications

AUG 2016 - SEP 2016

Internship cryptography

UNIVERSITY OF TECHNOLOGY, Graz, Austria
Institute of Analysis and Number Theory

OCT 2015 - JAN 2016

Undergraduate teaching assistant in real analysis



ACHIEVEMENTS

INGE St. research award in category best publications

2021

Poster prize

2019

NeurIPS workshop on biological and artificial RL
UNIQUE research centre

Google hashcode challenge, placed 111/3012

2018

Academic Excellence Scholarship
University of Technology Graz

2017

Academic Excellence Scholarship
University of Technology Graz

2017

Academic Excellence Scholarship
University of Technology Graz

2016

Academic Excellence Scholarship
University of Technology Graz

2014



SKILLS

Expert level knowledge of TensorFlow and Python

Analytical skills, broad understanding of diverse deep learning approaches

Applied skills in performing distributed training

Creative skills in visualization